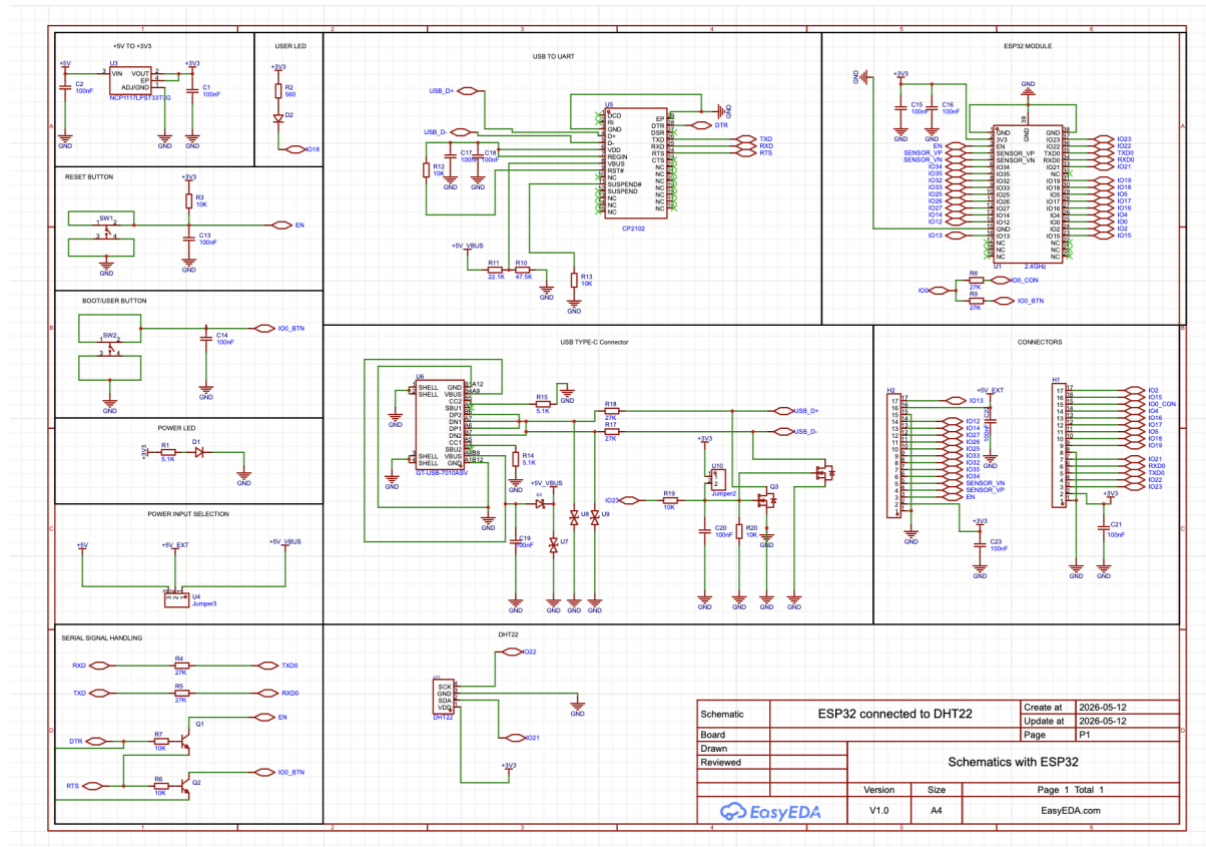


Embedded Systems & IoT April -July 2026

Lab Exercise: Schematics with ESP32

ESP32 connected to DHT22



Ref	Value / Part	Type
U1	ESP32 module	IC
U2	DHT22	IC
U3	NCP1117/LPS3373G	IC
U4	Jumper3	Connector
U5	CP2102	IC
U6	GT-USB-7010ASV	Connector
U7, U8, U9, U10	Diodes	Diode
D1	LED (power)	LED
D2	LED (user)	LED
Q1, Q2	NPN transistor	Transistor
Q3	Jumper2	Connector
SW1	Reset button	Switch
SW2	Boot/user button	Switch
H1, H2	Pin headers	Connector
R1	5.1 kΩ	Resistor
R2	560 Ω	Resistor
R3	10 kΩ	Resistor
R4, R5	27 kΩ	Resistor
R6, R7	10 kΩ	Resistor

R8, R9	27 k Ω	Resistor
R10	47.5 k Ω	Resistor
R11	22.1 k Ω	Resistor
R12	10 k Ω	Resistor
R13	10 k Ω	Resistor
R14, R15	5.1 k Ω	Resistor
R17, R18	27 k Ω	Resistor
R19	10 k Ω	Resistor
R20	10 k Ω	Resistor
C1	100 nF	Capacitor
C2	100 nF	Capacitor
C13	100 nF	Capacitor
C14	100 nF	Capacitor
C15, C16	100 nF	Capacitor
C17, C18	100 nF	Capacitor
C19	100 nF	Capacitor
C20	100 nF	Capacitor
C21, C23	100 nF	Capacitor